



SOLIDIRON TABLET 200 MG

PRODUCT DESCRIPTION

Clean, dry, uniform, round, biconvex, even edged tablets and brown in colour.

Each tablet contains: Ferrous Fumarate 200mg.

PHARMACODYNAMICS

Iron is an essential constituent of the body, and is necessary for haemoglobin formation and the oxidative processes of living tissues. Iron and iron salts should be given for the treatment or prophylaxis of iron deficiency anaemias. Preparations of iron are administered by mouth, by intramuscular or intravenous injection.

Soluble ferrous salts are most effective by mouth. Ferrous fumarate is an easily absorbed source of iron for replacement therapy. It is a salt of ferrous iron with an organic acid and is less irritant to the gastro-intestinal tract than salts with inorganic acids.

PHARMACOKINETICS

Absorption

Once in the stomach, the acid conditions of the gastric contents cause the dissociation of ferrous fumarate and ferrous ions are liberated. These ions are absorbed through the proximal portion of the duodenum.

The ferrous iron absorbed by the mucosal cells of the duodenum is oxidised to the ferric form, and this is bound to protein to form Ferritin.

Distribution

Ferritin in the mucosal cells releases iron into the blood, where it is bound to transferrin and is passed onto the iron stores in the liver, spleen, and bone marrow.

These stores constitute a reserve of iron for synthesis of haemoglobin, myoglobin, and iron containing enzymes.

Elimination

Iron is lost from the body through loss of cells i.e, urine, faeces, hair, skin, sputum, nails, sloughing of mucosal cells, and through blood loss.

Ferrous fumarate has the same pattern of absorption and excretion as dietary iron.

INDICATIONS

Prophylaxis and treatment of Iron deficiency

Solidiron Tablet 200mg is indicated in the prevention and treatment of iron deficiency resulting from inadequate diet, malabsorption, pregnancy, parasitic infestation and/or blood loss.

RECOMMENDED DOSE

Adult

Prophylactic - Oral, 200mg daily.

Therapeutic - Oral, 200 mg three or four times a day, the dosage being adjusted gradually as needed and tolerated.

Paediatric Population:

Solidiron Tablet 200mg is not intended for the treatment of children.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS

Solidiron Tablet 200mg is contraindicated in patients with:

1. Hemochromatosis
2. Haemosiderosis
3. Anaemic conditions not due to iron deficiency
4. Hypersensitivity to ferrous fumarate
5. Paroxysmal nocturnal haemoglobinuria
6. Active peptic ulcer
7. Repeated blood transfusions
8. Inflammatory bowel disease including regional enteritis and ulcerative colitis, intestinal strictures and diverticulae.

WARNINGS AND PRECAUTIONS

Use of this medication should be carefully considered in patients with the following medical problems:

i. Post gastrectomy patients

Some post gastrectomy patients exhibit low iron absorption. Care is needed when treating patients with iron deficiency anaemia in patients with treated or controlled peptic ulceration. Duration of treatment of uncomplicated iron deficiency anaemia should not usually exceed 6 months (or 3 months after reversal of the anaemia has been achieved).

ii. Microcytic anaemia

Since anaemia due to the combination of iron and vitamin B12 or folate deficiencies in the form may be microcytic, patients with resistant to iron therapy microcytic anaemia should be tested for vitamin B12 or folate deficiency

iii. Paediatric population

This medication should be kept out of the reach and sight of children as overdose may be fatal.

iv. Iron preparations should be used with caution in patients with erythropoietic protoporphyria

v. Iron preparations colour the faeces black, which may interfere with tests used for detection of occult blood in the stools

vi. This medicine contains less than 1 mmol sodium (23 mg) per dosage, that is to say essentially 'sodium-free'.

INTERACTIONS WITH OTHER MEDICAMENTS

Be alert of the possible drug interactions when Solidiron Tablet 200mg is taken together with the following:

- i. Drugs used for treating infections, chloramphenicol, tetracyclines, ciprofloxacin, levofloxacin, moxifloxacin, ofloxacin, norfloxacin, and cefdinir where the absorption of the drugs are reduced by ferrous fumarate. Chloramphenicol delays plasma ferrous fumarate clearance, incorporation of ferrous fumarate into red blood cells and interferes with erythropoiesis. Absorption of both ferrous fumarate and antibiotic may be reduced if ferrous fumarate is given with tetracycline.
- ii. Medicines containing zinc, calcium or magnesium. Absorption of ferrous fumarate is reduced by calcium salts and magnesium salts.
- iii. Levothyroxine (thyroxine), used for thyroid gland problems. Ferrous fumarate should be given at least two hours apart.
- iv. Medicines used for a disease known as 'Parkinson's disease' (e.g. levodopa, entacapone).
- v. Methyldopa, used for increase in blood pressure. Concomitant use of iron with methyldopa may reduce the hypotensive effect of methyldopa.
- vi. Penicillamine used in the treatment of rheumatoid arthritis.
- vii. Some inhibition of iron absorption may occur if it's taken with cholestyramine, neomycin, antacids and proton pump inhibitors which reduce stomach acid and in the presence of food (e.g. tea, coffee, wholegrain cereals, eggs and milk). Bicarbonates, carbonates, oxalates or phosphates may impair the absorption of iron by the formation of insoluble complexes.
- viii. Medicines for bone diseases (bisphosphonates).
- ix. Trientine use for high copper levels in blood reduces the absorption of ferrous fumarate.
- x. Avoid concomitant use of ferrous fumarate with dimercaprol as toxic complexes may form.

xi. Mycophenolate, used during change of organs (transplant).

xii. Concurrent use of Vitamin E with iron supplements may impair hematologic response in patients with iron deficiency anemia; large doses of iron may increase daily requirements of Vitamin E; observation of patients receiving both is recommended.

xiii. Iron absorption may be increased by ascorbic acid or citric acid

xiv. Iron compounds impair the bioavailability of carbidopa.

xv. Concurrent administration of antacids may reduce absorption of iron.

xvi. Iron possibly reduces the absorption of eltrombopag (give at least 4 hours apart).

xvii. Iron chelates with acetohydroxamic acid reducing the absorption of both.

PREGNANCY AND LACTATION

Pregnancy

Solidiron Tablet 200mg can be used during pregnancy if clinically indicated.

Lactation

No adverse effects have been shown in breastfed infants of treated mothers. Solidiron Tablet 200mg can be used during breast-feeding if clinically indicated.

ADVERSE EFFECTS/UNDESIRABLE EFFECTS

The common adverse effects are:

Gastrointestinal disorders

Nausea, vomiting, gastrointestinal irritation, diarrhoea, epigastric pain, darkening of stools, intestinal strictures and diverticular disease. Iron preparations taken orally can be constipating, particularly in older patients and occasionally lead to faecal impaction.

Immune system disorders

Hypersensitivity reaction such as urticaria, rash, pruritus, oedema and photosensitivity.

Cardiovascular disorders

Ischemic heart disease.

Respiratory disorders

Difficulty for breathing.

Effects on growth

Iron supplementation in iron-replete children has been reported to adversely affect their growth.

If side-effects occur, the dose may be reduced; alternatively, another iron salt may be used but an improvement in tolerance may simply be a result of a lower content of elemental iron

SYMPTOMS AND MANAGEMENT OF OVERDOSE

Symptoms

Early Signs

Ingestion of 20 mg/kg elemental iron is potentially toxic and 200-250 mg/kg is potentially fatal. No single method of assessment is entirely satisfactory - clinical features as well as laboratory analysis must be taken into account. The serum iron taken at about 4 hours after ingestion is the best laboratory measure of severity. Each Iron Tablet 200 mg contains approximately 66 mg of elemental iron.

Serum Iron	Severity
<3 mg/L (55 micromol/L)	Mild toxicity
3-5mg/L (55-90 micromol/L)	Moderate toxicity
>5 mg/L (90 micromol/L)	Severe toxicity

Early signs and symptoms include nausea, vomiting, abdominal pain, haematemesis, rectal bleeding, lethargy, circulatory collapse, hyperglycaemia, metabolic acidosis and diarrhoea. The vomit and stools may be grey or black. In mild cases, early features improve but in more serious cases there may be evidence of hypoperfusion (cool peripheries and hypotension), metabolic acidosis and systemic toxicity. In serious cases there can be recurrence of vomiting and gastrointestinal bleeding, 12 hours after ingestion. Shock can result from hypovolaemia or direct cardiotoxicity. Evidence of hepatocellular necrosis appears at this stage with jaundice, bleeding, hypoglycaemia, encephalopathy and positive anion gap metabolic acidosis. Poor tissue perfusion may lead to renal failure. Rarely, gastric scarring causing stricture or pyloric stenosis (alone or in combination) may lead to partial or complete bowel obstruction 2-5 weeks after ingestion. Other symptoms in severe cases are coma, hypothermia, pulmonary oedema, diffuse vascular congestion, coagulopathy, convulsions, toxic encephalitis and central nervous system damage.

Management

Supportive and symptomatic measures include ensuring a clear airway, monitor cardiac rhythm, BP and urine output, establishing IV access and administering sufficient fluids to

ensure adequate hydration. Consider whole bowel irrigation. If metabolic acidosis persists despite correction of hypoxia and adequate fluid resuscitation, an initial dose of 50 mmol sodium bicarbonate may be given and repeated as necessary, for adults guided by arterial blood gas monitoring (aim for a pH of 7.4). Consider the use of desferrioxamine if the patient is symptomatic (other than nausea), serum iron concentration is between 3-5 mg/L (55-90 micromol/L) and still rising. Haemodialysis does not remove iron effectively but should be considered on a supportive basis for acute renal failure as this will facilitate removal of the iron-desferrioxamine complex.

EFFECTS AND ABILITY TO DRIVE AND USE MACHINE

No known effects have been reported.

INSTRUCTIONS FOR USE

- 1) This medicine is best taken on empty stomach, 1 hour before or 2 hours after meals for better absorption; but may be taken with food to lessen possibility of stomach upset.
- 2) Avoid regular use of large amounts of iron supplements several times daily for more than six months unless approved by physician.
- 3) Any sign of overdose must be immediately reported to physician.

Note: It is common that dust will be present on the tablet due to nature of the tablet itself. However, this does not affect the Quality, Efficacy and Safety of the product.

STORAGE CONDITIONS

Store below 30°C in a cool dry place, protect from light and moisture.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PACKING/PACK SIZES

Blister Pack of 10's (3 Strips of 10 Tablets).

REGISTRATION NO: MAL20020030XZ

KEEP MEDICINES OUT OF REACH OF CHILDREN
JAUHI UBATAN DARI KANAK-KANAK

For further information, please consult your doctor or your pharmacist

Revision Number: 08

Revision Date: 31 Jul 2023

Lt-089.08 PRODUCT REGISTRATION AND MANUFACTURER

IDAMAN PHARMA MANUFACTURING SDN BHD (200401023395)

LOT 24 & 25, JALAN PERUSAHAAN LAPAN,
BAKAR ARANG INDUSTRIAL ESTATE, 08000 SUNGAI PETANI,
KEDAH DARUL AMAN, MALAYSIA